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COMPRESSION SPRING DUAL-RATCHET WINCH FOR TIE-DOWN SYSTEMS

Abstract

A dual-ratchet winch attached to a plate, the dual-ratchet winch including: a ratchet mechanism coupled to and configured to rotate with respect to the plate, the ratchet mechanism including a pair of gears; a pawl including a pair of pawl tips and a locking hole, the pair of pawl tips configured to engage or disengage the pair of gears of the ratchet mechanism; a pair of wings to accept a pair of compression springs; and a pin assembly configured to keep the pair of pawl tips disengaged from the ratchet mechanism.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority under 35 U.S.C. § 119 (e) of co-pending U.S. Provisional Patent Application No. 63/560, 193, filed Mar. 1, 2024, entitled "Compression Spring Dual-Ratchet Winch for Tie-Down Systems." The disclosure of the above-referenced application is incorporated herein by reference.

BACKGROUND
Field of the Invention
[0002] The present disclosure relates generally to tie down systems and, more specifically, to compression spring dual-ratchet winch for the tie down systems.
Background
[0003] A car hauler trailer may have winches attached to the side to strap the vehicle(s) down to the trailer. A winch may include a ratchet mechanism to enable straps to be tightened and locked to keep the vehicle in place while the trailer is in transit.
[0004] FIG. **1** shows a simplified diagram of a ratchet mechanism **100** including a gear **110**, a pawl **120**, and a spring **130**. In FIG. **1**, the pawl **120** acts as a stopper for the gear **110** to prevent the movement of the gear **110** in a clockwise direction. In some industries and products, the pawl **120** may apply a force to the gear **110** due to the gravity. However, the ratchet mechanism **100** used for the car hauler trailer cannot rely solely on the force of gravity because the winch is not always in an orientation where gravity acts on the pawl **120** in the correct direction. Thus, the pawl **120** must be spring-loaded. The conventional ratchet mechanisms used in the car hauler trailers have used springs to keep the pawl engaged with the gear. However, the springs used in a configuration to keep the pawl engaged often fail and need to be replaced.
SUMMARY
[0005] The present disclosure describes a dual-ratchet winch for a car hauler trailer.
[0006] In one implementation, a dual-ratchet winch attached to a plate is disclosed. The dual-ratchet includes: a ratchet mechanism coupled to and configured to rotate with respect to the plate, the ratchet mechanism including a pair of gears; a pawl including a pair of pawl tips and a locking hole, the pair of pawl tips configured to engage or disengage the pair of gears of the ratchet mechanism; and a pin assembly configured to keep the pair of pawl tips disengaged from the ratchet mechanism.
[0007] In another implementation, a dual-ratchet winch for a car hauler trailer is disclosed. The dual-ratchet winch includes: a plate attached to a side of the car hauler trailer; a gear mechanism including a plurality of gears, the gear mechanism coupled to and configured to rotate with respect to the plate; and a pawl including a plurality of pawl tips configured to engage or disengage the plurality of gears by moving laterally forward or backward.
[0008] In a further implementation, a winch for a car hauler trailer is disclosed. The winch includes: a gear mechanism including at least one gear, the gear mechanism coupled to and configured to rotate with respect to a side of the car hauler trailer; a pawl including at least one pawl tip and a handle, wherein the at least one pawl tip is configured to disengage from the gear mechanism by moving the handle laterally away from the at least one gear; and a compression spring, wherein the at least one pawl tip is configured to engage the gear mechanism by the compression spring pushing the pawl laterally toward the at least one gear.
[0009] Other features and advantages of the present disclosure should be apparent from the present description which illustrates, by way of example, aspects of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The details of the present disclosure, both as to its structure and operation, may be gleaned in part by study of the appended drawings, in which like reference numerals refer to like parts, and in which:

[0011] FIG. **1** shows a simplified diagram of a ratchet mechanism including a gear, a pawl, and a spring;

[0012] FIG. **2**A is a diagram showing a car hauler trailer including winches attached to the side of the trailer in accordance with one implementation of the present disclosure;

[0013] FIG. **2**B is a detailed view of a winch shown in an inset diagram in accordance with one implementation of the present disclosure;

[0014] FIG. **3**A is an isometric view of a dual-ratchet winch coupled to a plate which is attached to a platform in accordance with one implementation of the present disclosure;

[0015] FIG. **3**B is a side view of a dual-ratchet winch in accordance with one implementation of the present disclosure;

[0016] FIG. **4**A is a detailed isometric view of a dual-ratchet winch in accordance with one implementation of the present disclosure;

[0017] FIG. **4**B is a top sectional view with the cutting plane positioned to view the dual-ratchet winch in further detail;

[0018] FIG. **5** is a detailed isometric view of the pawl in accordance with one implementation of the present disclosure;

[0019] FIG. **6** is a detailed isometric view of the pawl mounting plate in accordance with one implementation of the present disclosure; and

[0020] FIG. **7** is a detailed isometric view of the pin assembly mounted above the pawl mounting plate in accordance with one implementation of the present disclosure.

DETAILED DESCRIPTION

[0021] As stated above, conventional ratchet mechanisms used in the car hauler trailers have used springs to keep the pawl engaged with the gear. However, the springs used in the conventional ratchet mechanisms often fail and need to be replaced. To improve the design of the existing ratchet mechanism, certain implementations of the present disclosure provide for using pawl, mounting plate, and locking pin along with appropriately-placed springs to engage or disengage the winch gear assembly.

[0022] Accordingly, after reading this description it will become apparent how to implement the present disclosure in various implementations and applications. Although various implementations of the present disclosure will be described herein, it is understood that these implementations are presented by way of example only, and not limitation. As such, this detailed description of various implementations should not be construed to limit the scope or breadth of the present disclosure.

[0023] FIG. **2**A is a diagram showing a car hauler trailer **200** including winches **210-230**, **240** attached to the side of the trailer **200**. In one implementation, a winch (e.g., winch **240**) is coupled to a plate which is attached or is part of the side of the trailer **200**.

[0024] FIG. **2**B is a detailed view of a winch **240** shown in an inset diagram. The winch **240** is used to tighten and lock the strap **250** around the tire **252** of the vehicle **254** to keep the vehicle in place (i.e., in a tie-down system) while the trailer is in transit.

[0025] FIG. **3**A is an isometric view of a dual-ratchet winch **300** coupled to a plate **312** which is attached to a platform **310** in accordance with one implementation of the present disclosure. In the illustrated implementation of FIG. **3**, the dual-ratchet winch **300** also couples to a winch mandrel **320** including a slot **330** to slide a strap through it.

[0026] FIG. **3**B is a side view of a dual-ratchet winch **300** in accordance with one implementation of the present disclosure. In FIG. **3**, the dual-ratchet winch **300** includes a ratchet mechanism **340** and a pawl **350** to engage or disengage the ratchet mechanism **340**. In one implementation, the ratchet mechanism **340** is configured with two ratchets or gears (i.e., dual-ratchet or dual-gear). As will be shown in the subsequent figures, the pawl **350** is configured with two pawl tips and two pawl guides.

[0027] FIG. **4**A is a detailed isometric view of a dual-ratchet winch **400** in accordance with one implementation of the present disclosure. In the illustrated implementation of FIG. **4**A, parts of the winch **400** is shown in more detail. For example, the winch **400** includes a dual-ratchet mechanism **430** having two gears and a pawl **450** having a pair of pawl tips **420** and a pair of pawl guides **422** attached to it **450**. The pawl **450** also includes a T-grip handle **410** and a locking hole **412** (see FIG. **5** for details). Each pawl tip of the pair of pawl tips **420** engages and disengages with each gear of the dual-ratchet mechanism **430**. In FIG. **4**A, the winch **400** also includes a pawl mounting plate **460** (see FIG. **6** for details), a pair of compression springs **440**, and a locking pin **414** attached to a pin assembly (not shown here; see FIG. **7** for details).

[0028] FIG. **4**B is a top sectional view with the cutting plane positioned to view the dual-ratchet winch **400** in further detail. In the illustrated implementation of FIG. **4**B, the winch **400** includes the dual-ratchet mechanism **430** having two gears. The pawl **450** includes a pair of pawl tips **420** and a pair of pawl guides **422** attached to it **450**. The pawl **450** also includes a T-grip handle **410** and a locking hole **412** (see FIG. **5** for details). As illustrated in FIG. **4**B, each pawl tip of the pair of pawl tips **420** engages and disengages with each gear of the dual-ratchet mechanism **430**. In FIG. **4**B, the winch **400** also includes a pawl mounting plate **460** (see FIG. **6** for details), a pair of compression springs **440**, and a locking pin **414** attached to a pin assembly **470** (see FIG. **7** for details).

[0029] In FIG. **4**B, the T-grip handle **410** enables an operator to pull out the pawl **450** to disengage the pair of pawl tips **420** from the two gears of the dual-ratchet mechanism **430**. Also, when the operator pulls the T-grip handle **410** out to a disengaged position to disengage the pawl tips **420** from the dual-ratchet mechanism **430**, the locking pin **414** (shown in FIG. **4**A) drops into the locking hole **412** to keep the pawl tips **420** disengaged from the dual-ratchet mechanism **430**. Thus, by securely and completely disengaging the pawl tips **420** from the ratchet mechanism **430**, the ratchets or gears can be rotated freely (e.g., by inserting a tie-down bar in the center holes **480**) to rotate the ratchet mechanism **430** with respect to the plate **312** to align or loosen the straps (e.g., strap **150** in FIG. **1**B). To re-engage the pawl tips **420** on the dual-ratchet mechanism **430**, the operator can pull up on the pin assembly **470** (see FIG. **7** for details) to lift the locking pin **414** out of the locking hole **412** and allow the pair of pawl tips **420** to move forward and engage the ratchets or gears of the dual-ratchet mechanism **430**. In one implementation, the compression springs **440** are configured to push the pawl **450** forward to allow the pair of pawl tips **420** to engage the ratchets or gears of the dual-ratchet mechanism **430**.

[0030] As shown in FIGS. **4**A and **4**B, a dual-gear and dual-pawl configuration of the winch **400** provides a second contact point for added rigidity and promotes a geometry that is wide enough for the T-grip handle **410** to be integrated into the pawl design. Since the T-grip handle **410** is ergonomic, it enables the user to pull the pawl back with one hand and operate the ratchet gear using a tie-down bar with the other hand. Further, the force exerted by the T-grip handle **410** on the dual-ratchet winch **400** is borne by the compression spring **440** (designed to be inexpensive and widely available) in the engaged configuration, which may incur less failures.

[0031] FIG. **5** is a detailed isometric view of the pawl **450** in accordance with one implementation of the present disclosure. In the illustrated implementation of FIG. **5**, the pawl **450** includes the T-grip handle **410**, the pair of pawl tips **420**, the pair of pawl guides **422**, and the locking hole **412**. As described above, when the operator pulls the T-grip handle **410** out to a disengaged position to disengage the pawl tips **420** from the dual-ratchet mechanism **430**, the locking pin **414** drops into the locking hole **412** to keep the pawl tips **420** disengaged from the dual-ratchet mechanism **430**. In one implementation, the pawl **450** also includes a pair of wings **424** with each wing configured to accept a compression spring **440**.

[0032] FIG. **6** is a detailed isometric view of the pawl mounting plate **460** in accordance with one implementation of the present disclosure. In the illustrated implementation of FIG. **6**, the pawl mounting plate **460** includes a slot **600** through which the pawl **450** is mounted and an opening **610** through which the locking pin **414** passes.

[0033] FIG. **7** is a detailed isometric view of the pin assembly **470** mounted above the pawl mounting plate **460** in accordance with one implementation of the present disclosure. In the illustrated implementation of FIG. **7**, when the operator pulls the T-grip handle **410** out to a disengaged position to disengage the pawl tips **420** from the dual-ratchet mechanism **430**, the locking pin **414** drops into the locking hole **412** to keep the pawl tips **420** disengaged. To re-engage the pawl tips **420** on the dual-ratchet mechanism **430**, the operator can pull up on the pin assembly **470** to lift the locking pin **414** out of the locking hole **412** and allow the pawl **450** (along with the pawl tips **420**) to move forward and engage the ratchets or gears of the dual-ratchet mechanism **430**.

[0034] In one implementation, the benefits of the dual-ratchet winch **240**, **200**, **300**, **400** described above: [0035] 1) enable distribution of the force and substantial lowering of the risk of a ratchet failure by using the dual-gear and dual-pawl configuration to generate two points of contact; [0036] 2) enable easy movement of the ratchets by using the T-grip handle to disengage and lock the pawls before rotating the ratchets to operate the strap; [0037] 3) enable easy disengagement of the pawls from the ratchets by using the locking pin automatically falling into the locking hole; and [0038] 4) lower maintenance cost by using inexpensive and widely available compression springs for keeping the pawls engaged.

[0039] In one particular implementation, a dual-ratchet winch attached to a plate is disclosed. The dual-ratchet winch includes: a ratchet mechanism coupled to and configured to rotate with respect to the plate, the ratchet mechanism including a pair of gears; a pawl including a pair of pawl tips and a locking hole, the pair of pawl tips configured to engage or disengage the pair of gears of the ratchet mechanism; and a pin assembly configured to keep the pair of pawl tips disengaged from the ratchet mechanism.

[0040] In one implementation, the pawl further includes a pair of pawl guides configured to guide a movement of the pawl to engage or disengage the ratchet mechanism. In one implementation, the pawl further includes a handle to pull the pawl out and to disengage the pair of pawl tips from the ratchet mechanism. In one implementation, the dual-ratchet winch further includes a pawl mounting plate including a slot through which the pawl is mounted. In one implementation, the pawl mounting plate further includes an opening through which the locking pin passes. In one implementation, the pawl further includes a pair of wings. In one implementation, the dual-ratchet winch further includes a pair of compression springs, each compression spring configured to insert into each wing of the pair of wings. In one implementation, the pin assembly includes a locking pin configured to drop into the locking hole to keep the pair of pawl tips disengaged from the ratchet mechanism. In one implementation, the dual-ratchet winch further includes a plurality of center holes into which a tie-down bar is inserted to rotate the ratchet mechanism with respect to the plate while the pair of pawl tips is disengaged from the ratchet mechanism.

[0041] In another particular implementation, a dual-ratchet winch for a car hauler trailer is disclosed. The dual-ratchet winch includes: a plate attached to a side of the car hauler trailer; a gear mechanism including a plurality of gears, the gear mechanism coupled to and configured to rotate with respect to the plate; and a pawl including a plurality of pawl tips configured to engage or disengage the plurality of gears by moving laterally forward or backward.

[0042] In one implementation, the pawl further includes a locking hole. In one implementation, the dual-ratchet winch further includes a pin assembly including a locking pin configured to drop into the locking hole to keep the plurality of pawl tips disengaged from the gear mechanism. In one implementation, the pawl further includes a plurality of pawl guides configured to guide a movement of the pawl to engage or disengage the gear mechanism. In one implementation, the dual-ratchet winch further includes a handle for the pawl, wherein the handle enables the pawl to be pulled out and disengage the plurality of pawl tips from the gear mechanism. In one implementation, the dual-ratchet winch further includes a pawl mounting plate including a slot through which the pawl is mounted. In one implementation, the pawl mounting plate further includes an opening through which the locking pin passes. In one implementation, the dual-ratchet winch further includes a plurality of compression springs coupled to the pawl.

[0043] In another particular implementation, a winch for a car hauler trailer is disclosed. The winch includes: a gear mechanism including at least one gear, the gear mechanism coupled to and configured to rotate with respect to a side of the car hauler trailer; a pawl including at least one pawl tip and a handle, wherein the at least one pawl tip is configured to disengage from the gear mechanism by moving the handle laterally away from the at least one gear; and a compression spring, wherein the at least one pawl tip is configured to engage the gear mechanism by the compression spring pushing the pawl laterally toward the at least one gear.

[0044] In one implementation, the pawl further includes a locking hole. In one implementation, the winch further includes a pin assembly including a locking pin configured to drop into the locking hole to keep the plurality of pawl tips disengaged from the gear mechanism.

[0045] The disclosed implementations are provided to enable any person skilled in the art to make or use the disclosure. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles described herein can be applied to other implementations without departing from the spirit or scope of the disclosure. For example, although the ratchet mechanism and the pawl are described as being configured with two gears and tips, the ratchet mechanism and the pawl may be configured with any number of gears and tips. Thus, it is to be understood that the description presented herein shows implementations representative of the subject matter which is broadly contemplated by the present disclosure. All features of each implementation are not necessarily required in a particular implementation. Additional variations and implementations are also possible. Accordingly, the present disclosure is not limited only to the specific implementations describe above.

Claims

1. A dual-ratchet winch attached to a plate, the dual-ratchet winch comprising: a ratchet mechanism coupled to and configured to rotate with respect to the plate, the ratchet mechanism including a pair of gears; a pawl including a pair of pawl tips and a locking hole, the pair of pawl tips configured to engage or disengage the pair of gears of the ratchet mechanism; and a pin assembly configured to keep the pair of pawl tips disengaged from the ratchet mechanism.

2. The dual-ratchet winch of claim 1, wherein the pawl further includes a pair of pawl guides configured to guide a movement of the pawl to engage or disengage the ratchet mechanism.

3. The dual-ratchet winch of claim 1, wherein the pawl further includes a handle to pull the pawl out and to disengage the pair of pawl tips from the ratchet mechanism.

4. The dual-ratchet winch of claim 1, further comprising a pawl mounting plate including a slot through which the pawl is mounted.

5. The dual-ratchet winch of claim 4, wherein the pawl mounting plate further includes an opening through which the locking pin passes.

6. The dual-ratchet winch of claim 1, wherein the pawl further includes a pair of wings.

7. The dual-ratchet winch of claim 6, further comprising a pair of compression springs, each compression spring configured to insert into each wing of the pair of wings.

8. The dual-ratchet winch of claim 1, wherein the pin assembly includes a locking pin configured to drop into the locking hole to keep the pair of pawl tips disengaged from the ratchet mechanism.

9. The dual-ratchet winch of claim 8, further comprising a plurality of center holes into which a tie-down bar is inserted to rotate the ratchet mechanism with respect to the plate while the pair of pawl tips is disengaged from the ratchet mechanism.

10. A dual-ratchet winch for a car hauler trailer, the dual-ratchet winch comprising: a plate attached to a side of the car hauler trailer; a gear mechanism including a plurality of gears, the gear mechanism coupled to and configured to rotate with respect to the plate; and a pawl including a plurality of pawl tips configured to engage or disengage the plurality of gears by moving laterally forward or backward.

11. The dual-ratchet winch of claim 10, wherein the pawl further includes a locking hole.

12. The dual-ratchet winch of claim 11, further comprising a pin assembly including a locking pin configured to drop into the locking hole to keep the plurality of pawl tips disengaged from the gear mechanism.

13. The dual-ratchet winch of claim 10, wherein the pawl further includes a plurality of pawl guides configured to guide a movement of the pawl to engage or disengage the gear mechanism.

14. The dual-ratchet winch of claim 10, further comprising a handle for the pawl, wherein the handle enables the pawl to be pulled out and disengage the plurality of pawl tips from the gear mechanism.

15. The dual-ratchet winch of claim 10, further comprising a pawl mounting plate including a slot through which the pawl is mounted.

16. The dual-ratchet winch of claim 15, wherein the pawl mounting plate further includes an opening through which the locking pin passes.

17. The dual-ratchet winch of claim 10, further comprising a plurality of compression springs coupled to the pawl.

18. A winch for a car hauler trailer, the winch comprising: a gear mechanism including at least one gear, the gear mechanism coupled to and configured to rotate with respect to a side of the car hauler trailer; a pawl including at least one pawl tip and a handle, wherein the at least one pawl tip is configured to disengage from the gear mechanism by moving the handle laterally away from the at least one gear; and a compression spring, wherein the at least one pawl tip is configured to engage the gear mechanism by the compression spring pushing the pawl laterally toward the at least one gear.

19. The winch of claim 18, wherein the pawl further includes a locking hole.

20. The winch of claim 19, further comprising a pin assembly including a locking pin configured to drop into the locking hole to keep the plurality of pawl tips disengaged from the gear mechanism.